

EcoDrive Intelligence

Smart PLC - HMI - VFD - Energy Management System

for Real-Time Industrial Energy Optimization

Schneider Electric HMI Hackathon — Round 1 Proposed Solution

Team Name : <Your Team Name>

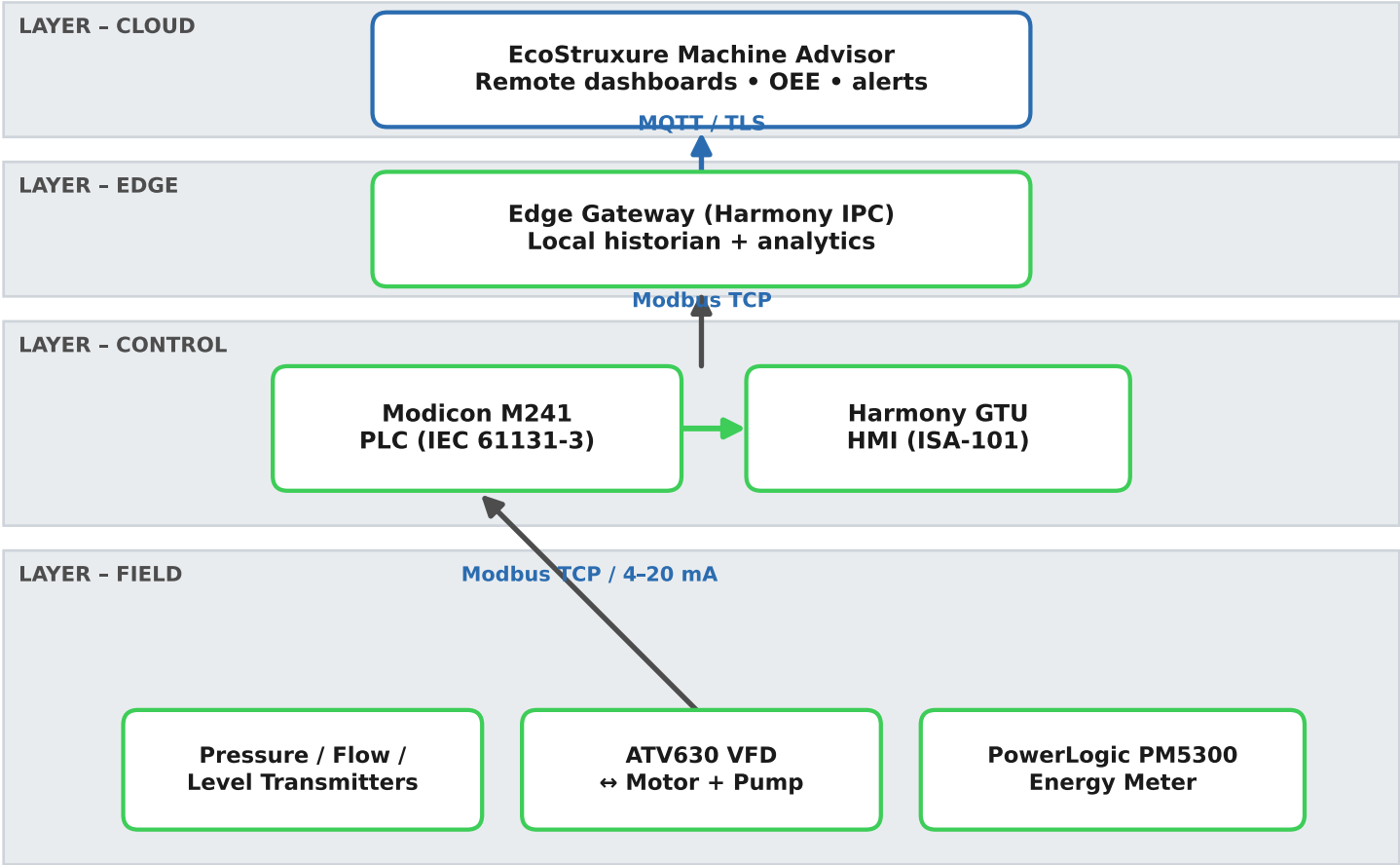
Member 1 : <Name>, <Dept>

Member 2 : <Name>, <Dept>

Member 3 : <Name>, <Dept>

Departments: EIE / ECE / CSE (interdisciplinary) • Batch 2027-2030

Architecture / Approach



Approach

- ▶ Use case: friction-dominated water pumping station (30 kW, variable demand).
- ▶ VFD speed control replaces valve throttling → cube-law savings ($P \propto N^3$).
- ▶ Savings calibrated to the real system curve (static vs. friction head) — honest, verifiable.
- ▶ Closed loop: SENSE → CONTROL → VISUALIZE → ANALYZE → OPTIMIZE → ACT → LEARN.
- ▶ Modbus TCP common layer — native to all four Schneider devices.

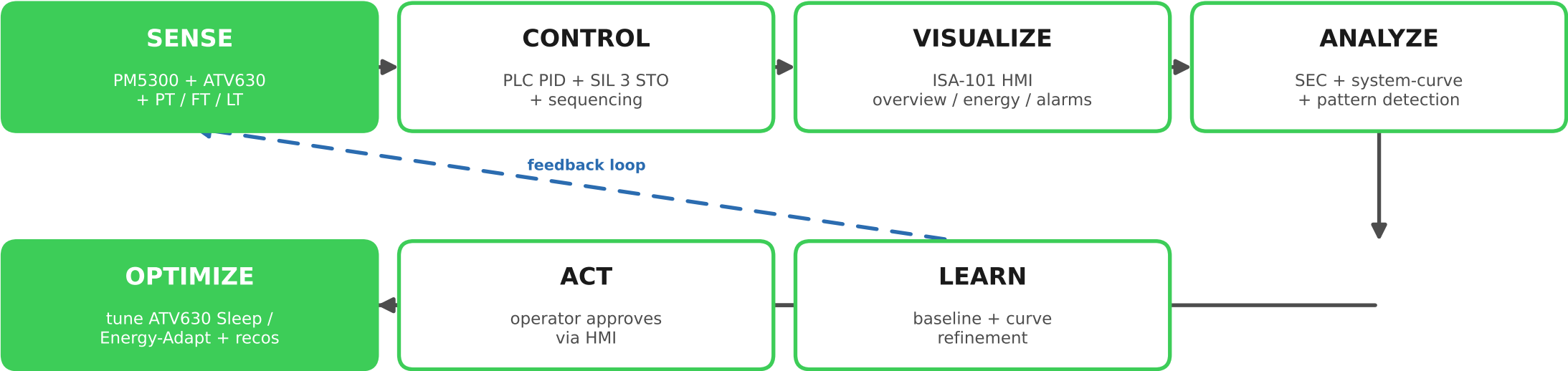
~46% Energy ↓

SEC 0.58→0.31 kWh/m³

₹4-6 L / year saved

Payback < 2 years

Overall Sequence Flow



PLC control sequence:



Covers all 7 objectives: Comms → Control → HMI → Energy Dashboard → Alarms (ISA-18.2) → Optimization → IoT

Tools Used

Schneider Hardware

- Modicon M241 — PLC
- Harmony GTU — HMI
- Altivar Process ATV630 — VFD
- (SIL 3 STO, Sleep Mode, embedded energy monitor)*
- PowerLogic PM5300 — meter
- Acti9 iEM3155 — sub-meter
- Pressure / Flow / Level Tx
- ConneXium managed switch

Software & Protocols

- EcoStruxure Machine Expert
- (IEC 61131-3 - ST/FBD/SFC/LD)*
- Vijeo Designer / Operator
- Terminal Expert (HMI)*
- EcoStruxure Machine Advisor
- (IoT / cloud)*
- Modbus TCP/RTU • EtherNet/IP
- MQTT/TLS • OPC UA • 4-20 mA
- Prototype: Node-RED • Grafana
- Raspberry Pi / ESP32

Standards Applied

- ISA-101 — High-Perf HMI
- ISA-18.2 / IEC 62682 — Alarms
- IEC 61131-3 — PLC languages
- IEC 61508 — Functional safety
- (Safe Torque Off)*
- IEEE 519 — Harmonics / THD
- ISO 50001 — Energy mgmt
- IEC 62443 — OT cybersecurity

Entire solution built on Schneider Electric's EcoStruxure ecosystem • demo-ready via open-source edge stack for Round 2